

INSTITUTE OF ACTUARIES OF INDIA

Subject SA7 - Investment and Finance

November 2024 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1)**i.** Types of investment guarantees:

- Guaranteed Vesting Benefit (GVB) in nominal terms: Vesting Benefit (VB) at the time of retirement is determined using Guaranteed Growth Rate (GGR) of X% say 7% on the contributions made by a subscriber during accumulation phase. This is equivalent to DBS. $VB = GVB$.
- GVB in real terms: similar to GVB in nominal terms but GGR is Y% + inflation index returns say 1% + inflation rate.
- Minimum Absolute Return (MAR): All contributions grow at MAR to give GVB. MAR could be 0% (=capital guarantee) or $MAR > 0\%$ say 3%. VB at retirement is higher of Actual Fund Value (AFV) and GVB;
 $VB = \text{Max} (GVB, AFV)$.

Shortfall = $\text{Max} (GVB - AFV, 0)$ and surplus = $\text{Max} (AFV - GVB, 0)$; Actual returns under MAR depends on actual fund performance. Any shortfall is made good by sponsor / PFMC or both in agreed proportion. Any surplus is credited to subscriber's account.

- Minimum Relative Return (MRR): similar to MAR, but contributions grow at returns linked to Government Bond (GB) yields.
For example, $MRR = X\% \text{ GB yield}$ say 50% of one year GB yield; or
 $MRR = \text{GB yield} - Y\%$, say ten year GB yield – 1.5%.
- The above guarantees could be applicable only on retirement (terminal) or on an on-going basis i.e. on death/ transfer/ partial withdrawals/ retirement.
On-going guarantee is more onerous.
- Another guarantee could be offered by a tie-up of PFMC with Annuity-Service-Provider (ASP) where ASP offers guaranteed annuity option i.e. the minimum rate of converting a lump sum in to annuity.

[4]

ii. Issues where ALM analysis is required

What kind of guarantee: absolute or relative rate of return guarantees can be reasonably provided by a PFMC given its capital and economic environment

Optimal guarantee reset period (Quarterly / half yearly / annual) that strikes a balance between subscriber interest and PFMC who offers a guaranteed CPS

Determine optimal asset allocation to money market instruments, GBs, Corporate debt and equity; can be done in following ways:

- Using Cash flow allocation strategy: Set allocation proportions of new contributions and asset income cash flows like coupons / redemptions due to maturities etc.
- Using Target allocation strategy: Rebalance portfolio at desired frequency as per target proportions to various asset classes
- Using Life cycle strategy: Rebalance portfolio at target proportions dependent on age of the subscriber; choice of investment option aggressive/ moderate/ conservative and term remaining to retirement.

Market consistent Valuation of the assets and liabilities of PFMC offering investment guarantees

Regular ALM reports on status of guarantees: whether guarantee is in-the-money i.e. it is biting or out-of-the –money

Stress testing / scenario analysis to assess level of shortfall is within tolerable limits

Stress testing / scenario analysis for individual guaranteed CPS's capital assessments and transfers / reflection of liabilities

Assessing solvency Capital / risk based capital requirement for PFMC and the ongoing need for capital for all CPSs offered with or without guarantees.

Determine methods and appropriate bases for assessing the on-going solvency of PFMC, considering the capital requirements

Set a basis for measuring and monitoring key metrics through ALM analyses, including stochastic approaches, for management of guaranteed CPSs

Set metrics for internal controls and procedures necessary to run and implement investment guarantees and enable PFMC in proper financial reporting

Develop appropriate ALM models for guaranteed CPS to determine

- Cost of Guarantees (CoG) offered
- Investment Guarantee Charge (IGC): fee /charged for the guarantee offered.
- Investment Guarantee Reserve (IGR): amount set aside as reserves to meet future expected shortfalls
- For on-going guarantees, on valuation basis, set floor to IGR = Max (current short fall, previous IGR with interest + current IGC)

Managing terminal guarantees is relatively simple but for on-going guarantees, determine:

- waiting period for the guarantees to be applicable like 3/5/7/10 years, if a waiting period is required
- minimum target allocation to stable assets (e.g. >70%) like money market instruments/ bonds
- cap on target allocation to volatile assets in aggregate (e.g. <30%) like equities, real estate
- exit loads/ penalty with respect to Market Value Adjustment (MVA) for pre-retirement exits other than death like transfers / partial withdrawals

For ASPs offering guaranteed annuity option,

- determine the minimum guaranteed rate of converting a lump sum in to annuity
- develop appropriate hedging strategies to mitigate risks

[13]

iii. Common risks faced by a PFMC:

Market risk is the risk relating to changes in the value of the portfolio due to movements in the market value of the assets held. It can be measured / monitored using Value at Risk.

Credit risk is the risk that counterparties to an agreement will be unable or unwilling to fulfill their obligations. It can be measured / monitored using credit scores or ratings.

Liquidity risk is the risk of not having sufficient cash to meet operational needs at all times. Identifying and measuring liquidity risk include the cash budgeting / short-term financial planning techniques, gap analysis and duration analysis.

Operational risk is the risk of loss due to fraud or mismanagement within the organisation; failed or inadequate people, processes or systems and external events. It is difficult to quantify this risk.

Relative performance risk is the risk of under-performing comparable PFMCs. This risk can be reduced by mimicking the competitors' actions but it is hard to find the required information on time to do so.

Product / business risk: Contributions / charges collected from subscribers are not sufficient to cover the product development and other regular expenses due to more than expected expense inflation and / or lower than expected sales volume.

Cyber risk: Risk of clients' data breaches / theft due to cyber crimes like ransom-ware attacks. This risk can be controlled by investing in technology architecture with firewalls and anti-virus or anti-malware software.

Regulatory / compliance risk: Non-compliance of existing and new regulations may lead to heavy penalties or closure to new business or even cancellation of licences to do business. This risk can be controlled by having a dedicated compliance team closely monitoring the developments in regulations.

[6]

iv. Stakeholders in a CPS are:

- PFMC
- Sponsor
- Subscribers
 - Opting guaranteed CPS
 - Opting non-guaranteed CPS
- ASP
- Government / Regulators

PFMC primarily faces market risk depending on the value of fund of guaranteed CPS changing adversely against its liabilities.

Sponsors also face market risk while paying / sharing the burden of shortfalls from terminal and on-going guarantees.

PFMC bears liquidity and market risk for on-going guarantees from subscribers':

- Partial withdrawals;
- Death; and
- Transfers.

Under extreme market scenarios shortfalls are huge and probability of withdrawals / transfers increases due to subscribers' anti-selection against PFMC.

For subscribers opting guaranteed CPS, returns may be lower than those not opting guarantees to the extent of IGC paid.

PFMC faces operational risks in maintaining books of accounts and needs to comply with the Pension Regulations on monitoring status of guarantees etc.; non-compliance may attract penalties.

Where a PFMC manages both guaranteed and non-guaranteed CPSs, there is a risk for subscribers opting not-guaranteed CPS that PFMC would do internal fund transfers of deteriorating risky illiquid assets (like equities during recession) from the guaranteed CPS funds to not-guaranteed CPS funds in exchange for fixed income assets.

Internal fund transfers are done at current market prices for controlling shortfalls in guaranteed CPS funds up to the extent target asset allocation of non-guaranteed CPS funds allow. This will result in increasing the market risk that is fully borne by the subscribers opting not-guaranteed CPSs.

ASP that offered guaranteed annuity option faces market risk when interest rates are low, expense inflation risk and longevity risk.

Regulators will be concerned of leaving loop holes in preparing or implementing regulations that PFMC could take advantage leading to loss for some subscribers. [7]

- v. PFMC can control market risk by regular ALM modelling and reporting shortfalls i.e. difference between market values of the CPS funds against the GVB.

If it seems that IGR will not be sufficient in future then propose capital injections in advance to the board/ senior management and also to the sponsors for sharing shortfalls.

PFMC should avoid offering on-going guarantees and offer only terminal guarantees to reduce liquidity and market risks arising from anti-selection of subscribers.

PFMC can control operational risks by establishing appropriate internal reporting and by separation of front office and back office functions. Also establish clearly the roles and responsibilities of each employee.

A subscriber should choose a PFMC that charge low rates as IGC.

Regulators could prescribe regulation on limiting internal fund transfers between guaranteed and non-guaranteed funds. Till then...

... a person should avoid subscribing to non-guaranteed CPS with a PFMC that also offers guaranteed CPS...

...if it is unavoidable (when all PFMCs started offering both) then select a PFMC that offers very small/ basic guarantees like capital guarantee.

ASP could purchase:

- swaptions to mitigate market risk;
- longevity swaps to mitigate longevity risk; and
- real estate/ equities or index linked bonds to mitigate expense inflation risk.

[5]

- vi. Possible indicative reports

PFMC should develop interface to provide relevant reports to various stakeholders like subscribers, sponsors, Banks, Custodians, and Regulators etc

Daily Report on:

- Net fund receipt for CPSs
- Investment report on purchases
- Discrepancy/confirmation report on net payout
- Report of CPS wise payout position
- Redemption report on sales

- Net Asset Values (NAVs)
- Investment management fees charged to CPSs

Monthly Reports on:

- Reconciliation of AUM and Units
- Details of Portfolio for CPSs being managed by PFMC
- Statement of Purchases and Sales effected by PFMC
- Fund Inflows and Outflows of CPSs being managed by PFMC
- Report on transactions in securities of Group Company/ PFMC associates
- Statement of downgraded investments held
- Statement of investments classified as default
- Statement of exposure limits for CPSs being managed by PFMC
- Statement of investments not eligible for investment management fees

Quarterly Reports on:

- Report transactions in securities by key personnel of PFMC in their own name
- Report transactions in securities with any of PFMC's associates
- Overview of portfolio positioning including evaluation of
 - Current economic conditions
 - Prospects for securities markets
 - Justification for the positions and transactions in the portfolio
 - Attribution of performance over last quarter (and year when applicable) on absolute basis as well as relative to the specific benchmark
 - Outlook for returns for the portfolio
- Report on status of investment guarantees
 - Statement on amount of Investment guarantee reserves (IGR) held
 - Statement on amount of current shortfall
 - Statement on amount of investment guarantee charges (IGC) collected

Half-yearly Reports on:

- Copy of half yearly unaudited accounts of CPSs as per CPS accounting regulations
- Any change in interests of Directors

Annual Reports on:

- Sponsor's regulatory licenses status and details of any changes in name or capitalization of PFMC or Sponsor(s)
- Statement showing the amount of interest accrued but not realized
- All service contracts executed by PFMC, including outsourced activity
- All service contracts such as for custody arrangements and transfer agency of the securities are executed in the interest of subscribers
- Summary of all activities and compliance with regulations, guidelines, circulars
- Copy of the Audited Annual Report and other information including details of investments as per the CPSs accounting regulations
- Copy of unaudited provisional financial statements (Balance Sheet, Revenue Account, notes and schedules of each CPS)
- Voting Report including the number of votes cast (for, against, or abstained)
- Statement of whether PFMC has complied with its obligation to exercise its voting rights in subscriber's interests only
- Statement on amount of shortfall covered / realized
- Statement of IGR > IGC
- Report on stress test to check IGR is sufficient to meet stressed shortfall

Ad-hoc Reports on:

- Bio-data of all its Directors along with their interest in other companies within few days of their appointment
- Conflicts of interest
- Any other information as determined by the Regulator

[15]

[50 Marks]**Solution 2:**

- i. Long term investors in stocks employing buy and hold strategy could enhance returns by applying tactical strategies like Covered Call (CC) and Protective Put (PP) using exchange traded European-style options. Both CC/PP can affect the holding period of a stock for tax purposes. Exercising them could attract capital gains tax.

PP requires two-part forecast. First, long term forecast must be bullish for buying (or holding) the stock. Second, short-term forecast is bearish for buying put option to limit risk.

Suppose a downward trending stock is believed to reverse upward. Applying PP is advantageous than using stop-loss order while acquiring the stock to limit risk if the predicted change in trend does not occur...

...as a stop-loss order is price sensitive and usually get triggered by a sharp fluctuation in the stock price before reversing trend. Long put is limited by time, not stock price.

Apply PP before earnings report of a stock held that could send stock price sharply in either direction. Investor will benefit if the report is positive, and it limits risk of a negative report.

Irrespective of the view, it is desirable to hedge the stock held partly or fully by purchasing PP, as views may go wrong and stock may crash any time...

...and if stock price rises, the investor participates fully, less insurance cost of PP.

Delta of stock plus put portfolio in 1:1 ratio is always positive since delta of stock = 1 and $-1 < \text{delta of put} < 0$. If view on the stock is highly bearish then hedge fully (portfolio delta say -0.1 to 0.1) otherwise hedge partly (portfolio delta say 0.4 to 0.8).

Purchase the far dated put options and roll over before one or two months remaining to expiry. This will avoid losses from theta decay that will be severe when closer to the expiry date.

If a particular stock put options are not available for trading or thinly traded with huge bid-offer spreads, then consider hedging using sector or market index put options... ...with appropriate hedge ratio considering beta of the stock.

When stock price falls, use the proceeds from PP to purchase more of the stock at lower levels provided the stock's fundamentals are still good...

...otherwise switch over to another best stock suggested by research team.

Roll over the PP at close to expiry of put options for desired level of hedging (delta) to hold the stock going forward...

...especially for ITM (In-the-money) puts to avoid crystallization of capital gains tax.

While selling call options in CC, pick near expiry since theta decay will favour most.

As long as stock price remains below strike price, a rolling CC will keep on generating regular premium income to enhance portfolio return.

If stock price rises above strike price by expiry, any upside above strike price is lost and capital gains tax will be crystallized...

...hence do not apply CC in the planning stage if view on the stock is highly bullish.

Close the CC already in-force even for loss at implementing stage if view on the stock turns to highly bullish...

...and consider CC again only after the momentum has halted.

Apply CC with strike price above few strikes of the target price by expiry i.e. at:

- deep OTM (Out-of-the-money) if view is mildly bullish to neutral
- OTM but closer to ATM (At-the-money) if view is mildly bearish to neutral.

Roll over the CC if at close to expiry call option becomes ITM and view is intact to avoid crystallization of capital gains tax.

While rolling over ITM CC ensure credit i.e. slide up the strike price of CC subject to Sale price of next month call-option > purchase price of expiring ITM call option.

Close the CC to book profits if call option has lost most of its value say 80% or 90% during any time within expiry...

...and roll over CC provided the view is intact.

An upward trending stock is believed to reverse downward in some near future. Applying CC is advantageous than using limit order when waiting for its price to rise to a certain threshold...

...as CC ensures selling at target price similar to a limit order but with extra income of premium.

[7]

- ii. PQR fund managers' significant portion of salary (variable salary 40%) is dependent on meeting the benchmark...

...and CC may sometimes underperform simple buy and hold strategy when CC expires in deep ITM...

...whereas for DEF managers variable salary at stake is 20%.

Even if CC expires slightly in ITM, the stock delivery will attract huge capital gains tax at inappropriate time...

...disturbing the tax plan for PQR clients as their stock holding periods can be as long as 30 years...

...whereas for DEF clients capital gains would be relatively low as maximum holding period is three years.

Majority of stocks held by PQR might be small / mid-cap stocks not in FnO (Future and Options) segment so cannot use exchange traded derivatives for CC...

...but could use PP for partial hedging through index put options.

DEF fund managers might have agreed due to over enthusiasm of trading in derivatives as they are young with average age 25 years...

...whereas PQR fund managers were reluctant to use covered call may be due to lack of enthusiasm as they are old with average age 55 years.

[4]

- iii. Long term investors in stocks employing buy and hold strategy could enhance returns by lending / pledging their stocks in return for an agreed fee on per share basis.

An investor holding stocks could lend stocks through SLBM (Security Lending and Borrowing Mechanism) to traders who wish to ...

... short-sell the stocks speculating stock price will fall...

...short-sell the stocks arbitraging the reverse cash futures spread i.e. futures trading at discount to cash price.

The risk of short-selling is borne by the borrowing trader and the investor would earn risk-free incremental return on the idle stocks lent.

The only risk for lender is default by borrower where the lender receives close-out credit (an estimated value of the shares) instead of shares leading to crystallization of capital gains tax. [2]

iv. Factors affecting recent high growth rate:

Recently the electric vehicle market in India saw a spike in sales. EVs are popular as cleaner mode of transportation and public understands need to reduce pollution and greenhouse gas emissions.

With rising petrol and diesel prices, EVs look attractive to customers in the long run.

EVs' lower operational and maintenance costs make them financially attractive alternative for long-term savings.

The main problem with EVs is high initial purchase cost due to high cost of battery.

Government incentives on EVs made them affordable through subsidies. These incentives drive consumers to choose EVs.

Government has also offered policy support to automakers like substantial tax breaks and simplified registration processes for EVs.

Indian automakers are utilizing this opportunity by producing a wider variety of electric models.

This diversity allows consumers to choose an EV that fits their needs and budget.

To promote sale of EVs, government announced Electric Mobility Promotion Scheme (EMPS) in 2024.

EMPS offers subsidy on EVs for example subsidy of up to INR 10,000 per two-wheeler,

Challenges for future growth rate:

In spite the subsidies; EVs are still more costly than petrol / diesel vehicles mainly due to cost of battery.

Batteries become dead and need to be replaced in few years. This increases the cost.

Range anxiety is the fear of running out of battery without access to a charging station... due to the scarcity of charging infrastructure, especially in rural and suburban areas.

Re-fuelling petrol / diesel vehicles takes few minutes but to re-charge battery might take hours.

Apparently it seems using EVs will substantially reduce pollution, but reality is: majority of electricity production is still dependent on burning coal.

During summers, few battery explosions happened while EVs were parked under sun.

Possible solutions to tackle the challenges:

Advancements in battery technology:

- develop batteries at lower cost by exploring other abundant raw materials / chemicals
- develop safer and more efficient batteries that charge faster and last longer to reduce range anxiety

Expanding charging infrastructure:

- The government and private sectors should work together to build a wider network of charging stations across the country
- This includes installing stations in various locations, from urban centres to remote areas, to ensure that EVs are a viable option for masses

Start-up of ancillary companies providing battery services:

- By owning and renting batteries to customers will reduce EV purchase cost
- Reduce waiting time at charging stations by swapping of charged batteries with low/no charge rented batteries on payment of fee for the charging
- Customers going on vacation could deposit rented batteries at any charging station to reduce rent cost
- Servicing batteries on a scheduled basis to check health of the batteries

Continued incentives and support by government:

- Enhancing governmental incentives such as EMPS will help in maintaining the momentum
- Do awareness programs / ad campaigns of using EVs as cleaner mode of transportation
- Promote green energy initiatives like hydro projects, solar cells, wind mills etc. to replace completely the dependency of electricity generation on coal
- Promote R&D (Research and Development) activities to develop new battery technologies that are safer and more efficient by providing
 - direct grants for R&D to research scholars
 - substantial tax breaks / deductions for longer periods of time to encourage companies engaged in R&D

Overall, a collaborative approach from government and private sector as stated above is essential to tackle the challenges and achieve desirable growth of the EV industry.

[12]

- v. There has been an explosion of interest in factor exposures and the launch of a large number of products based on many different types of factor. The expressions ‘factor-based investing’ and ‘smart beta’ are often used to capture this broad field.

Smart beta investing emphasizes the use of alternative index construction rules like equal weighted or any other factor weighted instead of traditional market cap-weighted indices e.g. Sensex or BSE500 or Nifty 50.

Categorisation according to size of a company and ‘value’ represented by the security remain the most significant factor exposures today...

...other key factors include alpha, volatility, liquidity, quality, momentum etc.

Smart beta strategy is based on single or combination of multiple factors and is implemented like typical passive index strategy.

It combines the benefits of active and passive investing strategies. The key difference between active and smart beta investing is that for active fund managers there is full discretion to hold or replace a stock...

...whereas for smart beta fund managers every inclusion or removal of a stock is set on rules based system...

...though rules for a smart beta fund may be slightly different as set by the fund manager than the rules of alternative index benchmark.

Smart beta focuses on capturing investment factors or market inefficiencies in a rules-based transparent manner.

Popularity of smart beta increased due to a desire for portfolio risk management and diversification along factor dimensions...

...aiming enhanced risk adjusted returns above market cap-weighted indices.

[4]

vi. Rules to create / maintain index benchmark for a smart beta fund:

Number of stocks in a smart beta benchmark index is set to be N (e.g. 30 or 50).

Stock selection is designed to provide exposure to high ranking factor (alpha) companies constituting BSE500 at the time of inception / rebalancing.

The method employs a factor driven selection criteria that aims to maximize factor (alpha) value while providing stability and tradability...

...by applying some filters to make the benchmark index investible and replicable.

Weights for each selected stock could be equal weighted or free float market-cap based or some function of the factor value / ranking.

The method should specify

- the base date
- frequency of rebalancing say quarterly or annually etc.
- the time period (quarter or half-year or year) of past data to be considered in assessing the factor (alpha) values

Identify eligible stocks from BSE500 index and apply some sanity filters like:

- To allow stability apply filters like
 - minimum market cap
 - maximum leverage
 - minimum ROE (Return on Equity)
 - positive EPS (Earnings Per Share) in past 'n' years etc.
- To allow liquidity / tradability apply filters like
 - minimum average daily turnover / traded volume
 - minimum trading frequency say 90% in last one year
 - maximum average impact cost
 - maximum average bid-offer spread etc.

Alpha of eligible stocks are calculated using one year trailing prices adjusted for corporate actions, if any.

Stocks are ranked in descending order of alphas and top N=30 will form the index at inception and at rebalancing buffer is applied to reduce frequent replacements.

To reduce frequent replacement of index constituents, a buffer of say 50% or 100% is applied at each rebalancing. For N=30, buffer of 50% (100%) means existing constituent will be retained provided it ranks in top 45 (60) by alpha values.

If weights are factor based then stocks with high values of factor (alpha) will get higher weight and weight is proportional to its value i.e. weight of i^{th} stock = (alpha of i^{th} stock) / (sum of alphas of all N constituent stocks)

[6]

Any cap say 10% or 20% on weight of individual stock is applied to allow diversification where weights may go beyond cap between rebalancing periods.

vii. Possible merit is “diversification” benefit ...

... Crypto returns are loosely correlated with returns on other assets ...

... Crypto currencies may be able to marginally improve portfolio returns without increasing portfolio risk.

Crypto currencies’ prices are highly volatile that could be seen favourably by speculative investors / traders ...

...But they have limited use and appeal for institutional investors.

The fluctuations in value of crypto currencies would disturb most investors who see their savings / wealth change materially overnight.

They can be used as a medium of exchange although their ‘non government’ nature has arguably made them more attractive for transactions that may not be 100% legal.

It is necessary to convert standard currency into a crypto ‘wallet’ and then to convert it back at some point ...

... This exposes investor to the risk that the company conducting this business goes bust or is exposed to fraud or cyber-attack ...

... For example, crypto platform WazirX reports assets worth \$ 230 million were stolen / hacked.

In most exchanges / platforms, Crypto currencies are priced in foreign currency like \$ USD and not in INR leading to currency risk in redemption of crypto wallet.

Many central banks have actively strived to discourage individuals and companies from using crypto, saying that the risks of fraud and volatility are too great and represent a systemic risk.

Other key demerits arise from adverse taxation of Crypto currencies in India:

In Indian Budget 2022-23, taxation of Virtual Digital Assets (VDAs) like Bitcoin was introduced ...

... Income from transfer of any VDA is taxed at 30% (plus cess) irrespective of whether the income is treated as capital gains or business income.

Except cost of acquisition of VDA, no deduction / expenditure or allowance is allowed while computing taxable income ...

... For example, infrastructure cost incurred on mining crypto assets will not be treated as cost of acquisition.

Loss from transfer of VDA cannot be set off against any other income ...

... Losses incurred from one digital currency cannot be set-off against income from another digital currency ...

... And losses from VDAs cannot be carried forward to offset gains in future years.

A buyer who owes a payment to the seller must subtract the TDS of 1% and forward it to the central government ...

... Only the balance amount will be paid to the seller.

For transactions on Indian exchanges: the exchange will automatically deduct the TDS and pay the balance to the seller.

viii.

The definition of VDAs covers crypto assets like Bitcoin, Non-Fungible Tokens (NFTs) and 'notified assets' and does not include crypto derivatives.

Moreover, 'notified assets' have not been specified by the government yet. Since crypto derivatives are not crypto assets, they would be taxed as normal derivative products, and VDA taxation may not apply to them.

Therefore, gains in crypto derivatives would be taxed as per normal tax slab, and losses could be carried forward to offset gains in future years.

However, if investors choose to take delivery of the contract i.e. physical settlement, then the gains would be taxed as VDAs.

Income tax is a highly technical and evolving law. As per current taxation laws there seems no tax evasion by these exchanges that offer a crypto-INR paired derivatives.

But taxation risks exist and the government can look to add crypto derivatives under 'notified assets', which brings them under VDA taxation that is very stringent.

Though the classification change may not be implemented retrospectively, there are other risks associated with dealing in crypto derivatives.

Crypto derivative contracts may trade with small trading volumes, low liquidity and wider bid-ask spreads. It would be difficult to execute trades at desired prices.

In addition, there is investment risk in derivatives where leverage can amplify returns and can also magnify losses; most of the time leading to loss of entire capital.

[3]

ix. 95% one week VaR using percentiles = $10,00,000 (1 - e^{(-3.5\%)}) = 34,395$

Assuming Log Normal (LN) distribution of returns

95% one week VaR = $10,00,000 (1 - e^{(0.31\% - 1.6449 (2.43\%))}) = 36,200$

Assuming LN distribution of returns and 52 weeks in a year

95% 1 yr VaR = $10,00,000 (1 - e^{(52 (0.31\%) - 1.6449 (52^{0.5}) (2.43\%))}) = 1,19,298$

[5]

[50 Marks]
